DAVID CHEN
Lead Machine Learning Researcher & NLP Specialist
Location: Boston, MA
GitHub: github.com/dchen-ai-research
Google Scholar: scholar.google.com/citations?user=dchen_nlp

PROFESSIONAL SUMMARY
Distinguished AI and Machine Learning Researcher with 7 years of specialized focus on Natural Language Processing, Tra

AREAS OF EXPERTISE & TECHNICAL SKILLS
- Core AI/ML: Natural Language Processing, Large Language Models, Token Classification, Entity Recognition, PyTorch, H
- Programming: Python, C++, CUDA, R, SQL
- Data Infrastructure: MongoDB, Apache Kafka, Spark, DuckDB
- Model Optimization: Quantization (AWQ, GPTQ), vLLM, TensorRT-LLM, Groq LPU acceleration

PROFESSIONAL EXPERIENCE

Principal NLP Research Scientist | CogniTech AI Labs, Boston, MA
September 2020 · Present
- Pioneered multi-modal schema extraction models processing scientific publications with 96.8% F1-score.
- Optimized sequence chunking and attention mechanisms, cutting memory footprint during inference by 40%.
- Led research collaboration with university partners on mitigating hallucinations in structured LLM extraction.

Senior Research Engineer | Quantix AI, Cambridge, MA
June 2017 · August 2020
- Developed proprietary text classification models for financial document risk auditing.
- Scaled training pipelines across multi-node GPU clusters using PyTorch Distributed and Slurm.

EDUCATION
Doctor of Philosophy (Ph.D.) in Electrical Engineering & Computer Science
Massachusetts Institute of Technology (MIT), Cambridge, MA (2013 · 2017)
Dissertation: "Efficient Context Compression for Deep Neural Language Understanding"

Bachelor of Science in Mathematics and Computer Science
Carnegie Mellon University, Pittsburgh, PA (2009 · 2013)

KEY PROJECTS
- OpenSchema-NER: Open-source Named Entity Recognition framework fine-tuned for dense academic texts.
- DeepDocParser: Layout-aware spatial token classification engine utilizing vision-augmented language models.

PUBLICATIONS & AWARDS
- Best Paper Award, EMNLP 2022: "Accelerated Structured Extraction from Semi-Structured Corpora"
- 6 Peer-Reviewed Papers in ACL, EMNLP, and NeurIPS workshops.